

A systematic review and meta-analysis of anti-predator mechanisms of eyespots: conspicuous pattern vs eye mimicry

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Ayumi Mizuno , Malgorzata Lagisz, Pietro Pollo, Yefeng Yang, Masayo Soma, Shinichi Nakagawa

Department of Biology, Faculty of Science, Hokkaido University, Sapporo, Hokkaido, Japan • Evolution & Ecology Research Centre, School of Biological, Earth and Environmental Sciences, The University of New South Wales, Sydney, New South Wales, Australia • Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology Graduate University, Onna, Okinawa, Japan

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Abstract

Eyespot patterns have evolved in many prey species. These patterns were traditionally explained by the eye mimicry hypothesis, which proposes that eyespots resembling vertebrate eyes function as predator avoidance. However, it is possible that eyespots are not the mimicry of eyes: according to the conspicuousness hypothesis, eyespots are just one form of vivid aposematic signals where only conspicuousness matters. To test these hypotheses and explore factors influencing predators' responses, we conducted a meta-analysis with 33 empirical papers focusing on bird responses to lepidopterans having conspicuous patterns (eyesspots and non-eyesspots). Supporting the latter hypothesis, the results showed no clear difference in predator avoidance efficacy between eyespots and non-eyesspots. When comparing geometric pattern characteristics, bigger pattern sizes and smaller numbers of patterns were more effective in preventing avian predation. This finding indicates that paired concentric patterns have weaker deterring effects than single ones. Taken together, our study supports the conspicuousness hypothesis more than the eye mimicry hypothesis. Due to the number and species coverage of published studies so far, the generalisability of our conclusion may be limited. The findings highlight that pattern conspicuousness is key to eliciting avian avoidance responses, shedding a different light on this classic example of signal evolution.

eLife assessment

This meta-analysis presents **valuable** findings that reexamine the function of butterfly eyespots in predator avoidance and report for conspicuousness over mimicry. The analysis is robust, but the evidence supporting the importance of conspicuousness is **incomplete** due to the limitations of the literature, and this debate would benefit from additional experiments that would strengthen these claims. This paper is of interest to evolutionary biologists and ecologists working on the evolution of morphology and predator-prey interactions.

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Background

Naturalists have long pondered the evolution and function of the many signals and cues animals use to communicate [1–9]. Visual signals, such as vibrant colours and contrasting patterns, have attracted more interest from researchers than other signals, likely because our species is visually oriented [1, 10, 11]. Eyespot patterns, characterised by concentric rings of different colours with a light outer ring and a dark centre [12], are well-known patterns believed to reduce predation. Although eyespots have been researched for a long time [12–15], researchers continue to debate why eyespots might deter predation.

Three hypotheses have been proposed to explain why eyespot patterns can contribute to prey survival (reviewed in [12, 14, 15]; **Fig. 1**). First, the eye mimicry hypothesis suggests that eyespots play a role in deterring predators from attacking prey and reducing predation risks by mimicking the eyes of vertebrates [16–18]. This hypothesis predicts that if the pattern has specific characteristics (e.g., eye-like shape) and is presented as a pair, predation avoidance will increase, assuming eyespots imitate potential predators. Second, the conspicuousness hypothesis posits that eyespots are simply conspicuous patterns that prevent attacks due to negative predator responses caused by sensory bias, neophobia, or sensory overload [12, 14]. The hypothesis states that the eye-like shape and patterns arranged in pairs have nothing to do with predator deterrence. Eyespots can act as an aposematic signal for potential predators. For example, if the size of the pattern (one of the measures of conspicuousness) increases, the avoidance effect will also increase. Third, the deflection hypothesis suggests that predator attacks should be directed towards eyespots to avoid damage to vital body parts [19–23]. The eye mimicry and conspicuousness hypotheses are usually applied to explain large eyespots, while the deflection hypothesis is used to interpret the function of small ones [12, 14, 15]. Although there seems to be little disagreement in the third hypothesis ([24–26], but see also [27]), why large eyespots can intimidate avian predators has been controversial [12, 14]. This is because while the eye mimicry and conspicuousness hypotheses are not mutually exclusive, the key mechanism that explains why predators react negatively to eyespots is clearly different.

Lepidopterans, such as butterflies and moths, have been the leading models for testing the eye mimicry and conspicuousness hypotheses. A typical empirical study has adult individuals, caterpillars, or their models as prey, with birds as predators (reviewed in [12, 14, 15]). According to the eye mimicry hypothesis, avian predators perceive the eyespots as the eyes of a potential enemy. For example, great tits (*Parus major*) showed more aversive responses to animated butterflies with a pair of large eyespots than those without, and such eyespots were more effective than modified, less mimetic, but equally contrasting patterns [28]. Although several studies have supported the eye mimicry hypothesis [e.g., 16, 28, 29], many

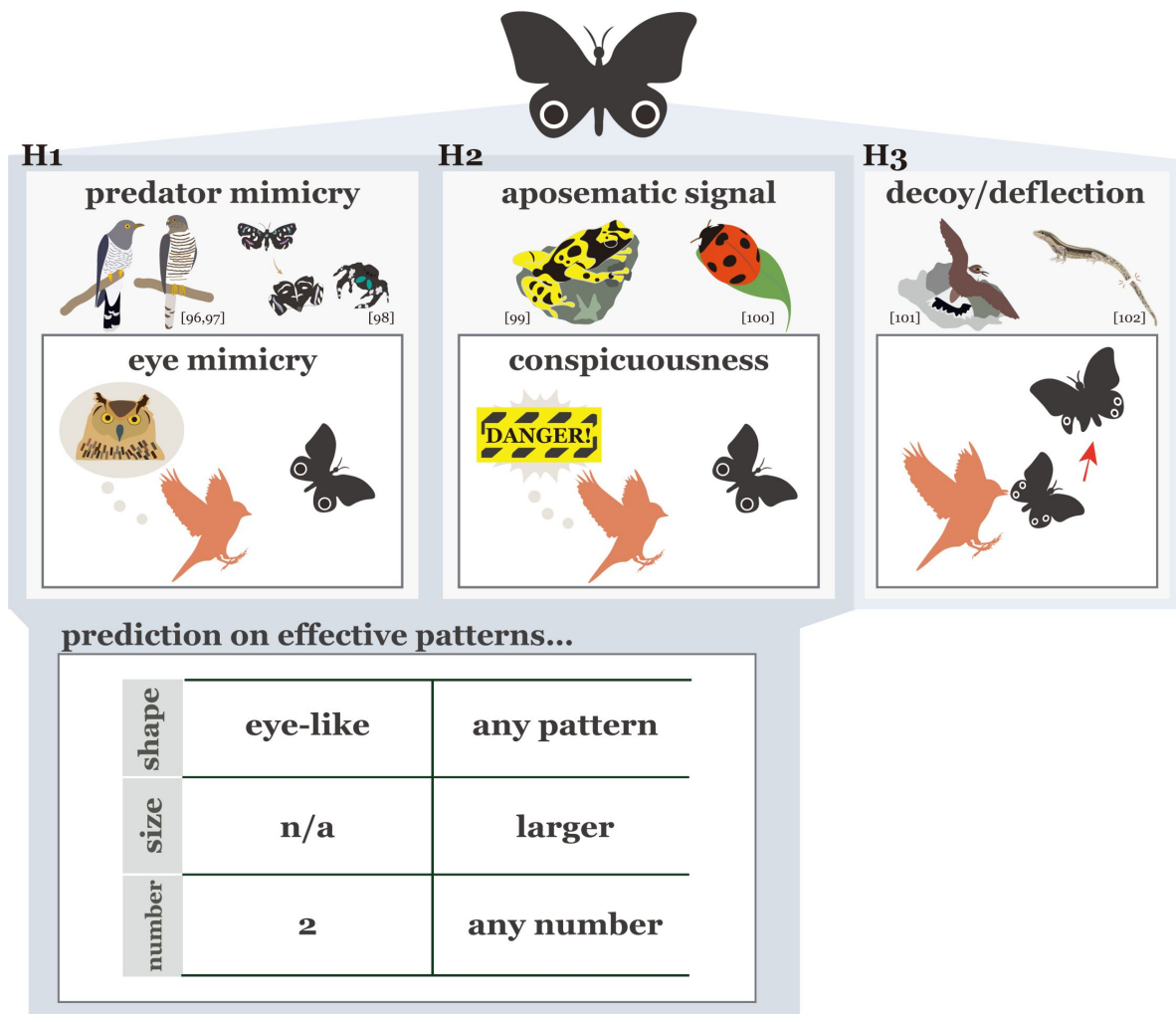


Figure 1

A visual summary of three hypotheses that explain the predation avoidance function of eyespot patterns and the predictions that can be derived from these two hypotheses.

conspicuous patterns other than eyespots, such as dots and stripes, likely deter attacks from predators as well [30–33]. Some field experiments with artificial prey have supported the conspicuousness hypothesis, demonstrating survival rates for both conspicuous (eyespot and non-eyespot) pattern prey stimuli were higher than control prey stimuli [30, 31, 34]. Such discrepancies might have arisen from differences in experimental design between studies, such as the size, number, and shape of the presented pattern stimuli or the bird species used as subjects in the experiments [12, 35]. However, there has been no systematic attempt to synthesise and compare earlier studies quantitatively.

Here, we conduct a systematic review with meta-analysis to synthesise empirical evidence on the intimidating effects of eyespots and the factors that contribute to predator avoidance responses towards them. To examine the two hypotheses above, we ask three interrelated questions. First, we examine whether conspicuous patterns, namely eyespots and non-eyespot patterns (i.e., conspicuous patterns other than eyespots), influence bird responses or prey survival in a manner that increases the success of predator avoidance. Second, we test whether pattern resemblance to eyes (eye-like shape) is the key to predator avoidance (which differentiates the eye mimicry hypothesis from the conspicuousness hypothesis). Third, we examine what factors promote bird response and increase prey survival, such as pattern size and the number of patterns (i.e., eyespots and non-eyespots; **Fig. 1**).

Materials and Methods

We preregistered our methods and planned analyses before data extraction and analysis in Open Science Framework (<https://osf.io/ymwvb>; [36]). We referenced and followed PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses; [37]) and PRISMA-EcoEvo (Preferred Reporting Items for Systematic reviews and Meta-analyses in Ecology and Evolutionary biology; [38]) for reporting this study (Table S1).

Search protocols

We used the PICO (Population, Intervention, Comparator, Outcome; Table 1) framework [39] to specify the scope of our research questions and to inform our literature searching and screening. We conducted a comprehensive literature search across multiple databases, including Scopus, ISI Web of Science, Google Scholar (for non-English studies), and Bielefeld Academic Search Engine (for unpublished theses; i.e., grey literature). We designed the search strings (see Table S2) to identify studies that used experimental methods to examine the effects of eyespot patterns on birds' predation behaviours. We did not set any temporal restrictions on the database searches. Additionally, we conducted backward and forward reference searches within the Scopus database using four key publications [12–15]. The strings were translated for searches in non-English languages, and search results were assessed by reviewers with expertise in the respective languages: AM for Japanese, ML for Polish and Russian, PP for Portuguese and Spanish, and YY for Simplified and Traditional Chinese. We limited Google Scholar searches to the top 100 results in each language, sorted by relevance. In cases of disagreement between the reviewers, discrepancies were discussed and resolved to reach a consensus. The screening process and results are shown in the PRISMA-like flowchart (**Fig. 2a**).

Eligibility criteria

We set specific criteria for including studies in our meta-analysis (according to our pre-registered protocol). Initial screening, including titles, abstracts, and keyword assessment for English-language bibliographic records, was conducted by AM and ML using Rayyan (<https://www.rayyan.ai>; [40]) following predefined inclusion criteria. Subsequently, AM and PP independently screened the full texts of studies that passed the initial screening. To be eligible, a study had to

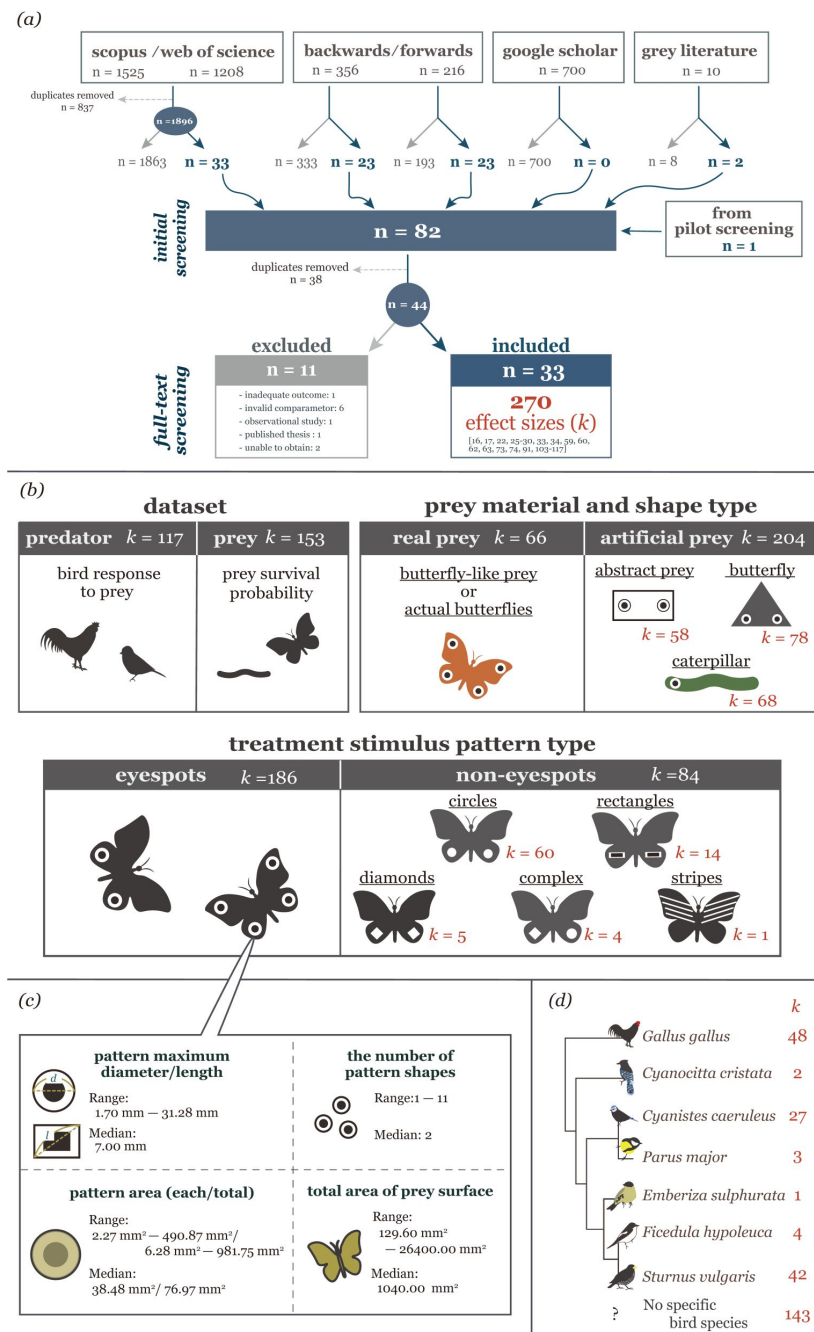


Figure 2

Overview of the dataset.

(a) shows a PRISMA-like flowchart of the systematic literature search for the meta-analysis. (b) and (c) give details of the main moderators examined in the meta-analysis. (d) provides the phylogenetic tree of bird species included in the meta-analysis, together with the sample sizes and number of effect sizes per species.

PICO	Description
Population	Birds as predators and butterflies, moths, caterpillars, and their models as prey
Intervention	Presenting eyespot or conspicuous pattern stimulus to birds
Comparator	Presenting stimulus that is neither eyespot nor conspicuous patterns
Outcome	Avian behavioural responses to eyespot or conspicuous pattern stimuli The probability of prey surviving or being attacked (for the stimuli)

Table 1

Descriptions of the population, Intervention, comparator, and outcome (PICO) used to define the scope of this study.

conduct experiments and provide data on bird behavioural responses or prey survival/attacked rates. We excluded studies solely involving non-avian predators, such as fish, insects, mammals, or other species. However, studies that included a mix of species from different taxonomic groups were allowed if the primary focus was on avian predation. In our analysis, we only considered research that presented both conspicuous and control (non-conspicuous) patterns as stimuli. We omitted studies using actual predator or human eyes as stimuli since we focused on understanding how eyespot patterns in butterflies and caterpillars, which are unlikely to resemble specific bird or vertebrate species eyes, affect predation avoidance [41]. We also excluded studies that used bright and contrasting patterns as control stimuli. Furthermore, we focused only on studies that used real or artificial butterflies, moths, caterpillars, or a piece of paper as prey or presented stimuli. We also did not consider research that only investigated avian physiological responses to conspicuous patterns. In addition, we did not include studies that only assessed whether prey with eyespots or conspicuous patterns were less likely to be attacked by birds, based on wing or body damage alone, without including control stimuli.

Data collection

We extracted four types of information from each study. First, we collected citation information, such as title, author name, and publication year. Second, we gathered the details of the presented stimuli used in each experiment within studies: type of control pattern (plain neutral-coloured or camouflaged), type of treatment pattern (eyespot or non-eyespot patterns), pattern area (mm²: area per shape comprising the pattern), total pattern area (mm²: when multiple patterns exist on the presented stimulus, it denotes the total area of all patterns; for stimuli with single eyespot or distinct pattern, the value equals the pattern area), linear size of the pattern (mm: e.g., maximum diameter or length of pattern), number of shapes in pattern, total area of prey surface (mm²: e.g., butterfly wings and caterpillar bodies), prey material type (i.e., whether a real butterfly or a complete imitation of a particular butterfly was used as prey), and prey shape type (a further subdivision of the former). For non-eyespot patterns, we also noted pattern shapes (e.g., circles, stripes, and triangles). In each study, bird responses to control and treatment pattern stimuli and prey survival/attacked rates when these patterns were present were reported. Bird responses contained a variety of measures, including the number of attacks and escape behaviours, latency to attack, latency to approach, and the proportion of birds attacking the presented stimuli. Henceforth, we refer to these measures and responses as ‘predator avoidance’. Third, we obtained data for calculating effect sizes (e.g., mean, standard deviation or standard error, and sample size of control and treatment group) from plots using WebPlotDigitizer 4.6.0 (<https://automeris.io/WebPlotDigitizer>), detailed tables, texts, or raw data. In survival analysis plots, we extracted data at the point in time when the difference between the ‘survival’ or ‘attacked’ rates of the intervention and comparison groups was greatest as outcomes. Study design (i.e., whether experiments were done independently or dependently between the control and treatment group) was also recorded. Fourth, we gathered predator and prey information, specifically, the study species (common English name and scientific name) and predator diet type. In some cases, studies did not use a specific bird species as a predator or a specific lepidopteran species as prey. We contacted authors when such information was ambiguous or missing. When the paper did not report the pattern area and diameter of the treatment stimulus or the presented stimulus surface area, AM calculated or measured them from available images using ImageJ v.1.53i [42].

The dataset was originally divided into two parts. The first part involved the data from presenting eyespot patterns to avian predators and directly observing their responses (predator dataset). The sample size or unit of analysis in this part was based on the number of individual avian predators. The second part involved the data from using real or artificial abstract butterflies, moths, or caterpillars with eyespots or non-eyespot patterns as stimuli or prey, and observing their survival/attacked probabilities in the field (prey dataset). The sample size or unit of analysis in this part was based on the number of real or artificial abstract prey. However, we also used the combined dataset that included both predator and prey datasets, as detailed in the “[Meta-analysis and meta-regressions](#)” and “[Publication bias](#)” sections.

Effect size calculation

To obtain the effect size point estimates and sampling variances, we used the natural logarithm of the response ratio (lnRR) between the means of the treatment and the treatment control stimulus groups [43–45]. Positive lnRR values indicate heightened aversion in birds and enhanced prey survival, while negative lnRR values signify diminished bird aversion and increased prey mortality. The point estimate and sampling variance (var) of lnRR can be then calculated in:

$$\ln RR = \ln\left(\frac{M_T}{M_C}\right) \quad (1)$$

$$\text{var}(\ln RR) = \frac{SD^2_T}{N_T M^2_T} + \frac{SD^2_C}{N_C M^2_C} - 2r \sqrt{\frac{SD^2_T}{N_T M^2_T}} \sqrt{\frac{SD^2_C}{N_C M^2_C}} \quad (2)$$

where M_T and M_C are mean responses of treatment and control groups (e.g., total frequency of attacking prey, latency of approach, or prey survivability), respectively. SD and N are (sample) standard deviations and sample size, respectively. The term, r is the correlation coefficient between responses of the two groups. Some of our eligible studies used the paired (dependent) study design where treatment and control samples originated from the same individuals, and sample sizes between the two groups were the same. None of these studies provided an estimate of r . Thus, when calculating our effect sizes, we assumed that this correlation was 0.5, which is conservative [46]. For the other studies that used independent study design, we set $r = 0$.

We note that our dataset included proportion (percentage) data (e.g., predator attack rate or prey survival probability), which are bounded at 0 (0%) and 1 (100%). Therefore, we transformed group means (M) and group standard deviations (SD) for proportion data using Equations (3) and (4) before applying (1) and (2) to calculate lnRR and the sampling variance:

$$f(M) = \arcsine(\sqrt{M}) \quad (3)$$

$$SD(f(M)) = \sqrt{\frac{SD^2}{4M(1-M)}} \quad (4)$$

where f indicates a function, in our case, the arcsine transformation. The standard deviation (SD) related to this transformation was derived using the delta method before calculating lnRR and the sampling variance [47]. We have also assumed that the standard deviation was $SD(f(M)) = 1/\sqrt{8}$ if SD was not available.

Meta-analysis and meta-regressions

We used the *rma.mv* function from the package *metafor* v.4.4.0 [48] in R v.4.3.1 [49] for our analyses. We started by fitting multilevel, mixed-effect meta-analytic models to the predator and prey datasets. These meta-analytic models explicitly incorporated random factors, Study ID, Cohort ID (groups of the same subjects), and Shared control ID (indicating effect sizes sharing control groups) [50] along with Observation ID, fitted by the above function [48]. The model for the predator dataset included Species ID and a correlation matrix related to phylogenetic relatedness for the species as random factors [51]. This is because we had data on the bird species used in the experiment in the predator dataset, and we needed to control for phylogenetic relationships between birds. We also quantified the total I^2 (a measure of heterogeneity not attributed to sampling error [52]) and how much each random factor was explained (partial I^2),

calculated by the *i2_ml* function from the package *orchaRd* v.2.0.0 [53]. After running both meta-analytical models, we found that phylogeny and Species ID did not need to be controlled for in the predator dataset, as their partial I^2 were zero ($I^2 = 0.00\%$). That is, these factors explained little heterogeneity between effect sizes.

Therefore, we merged predator and prey datasets (i.e., full dataset) without considering phylogenetic information and used them for the following models. We had, as random effects, Study ID, Cohort ID, Shared control ID, and Observation ID for our meta-analytic model using the full dataset. The Cohort ID and Shared control ID were removed from our subsequent meta-regressions because they both explained little heterogeneity (both partial $I^2 < 0.001\%$). This intercept-only (meta-analytic) model tested the conspicuous patterns (eyespot and non-eyespot) that affected predator avoidance (i.e., our first question).

Next, we tested whether eyespots and non-eyespot patterns differ in the magnitude and direction of the effect of elicited bird predator avoidance and what factors contribute to the deterring effects of conspicuous patterns. We performed uni-moderator meta-regression models with each of eight moderators: treatment stimulus pattern types (eyespot vs. non-eyespot), pattern area, the number of pattern shapes, prey material type, maximum pattern diameter/length, total pattern area, total area of prey surface, and prey shape type. We also ran a multi-moderator meta-regression model, including the first four of the eight variables mentioned in the uni-moderators, due to moderator correlations. We used log-transformed data for pattern area, total pattern area, total area of prey surface, and pattern maximum diameter/length in our analysis to normalise these moderators. We created all result plots in the *orchard_plot* and *bubble_plot* functions from the package *orchaRd* [53].

Publication bias

We used three approaches to assess the presence of publication bias in our study. First, we visually assessed the funnel plot asymmetry by examining the residuals from a meta-analytic model, which included all the random factors utilised in our study. These residuals were plotted against the precision of the effect sizes. Secondly, we performed an alternative method to Egger's regression. This method used the inverse of the effective sample size as a moderator within a multilevel meta-analytic model [54]. Third, we examined the possibility of time-lag bias by including publication year as a moderator in our multilevel meta-analytic model. Uni-moderator models were run for each inverse of the effective sample size and publication year, and a multi-moderator model was carried out with the full model including both inverse of the effective sample size and publication year as moderators.

Additions and deviations

We made two changes to the pre-registration: the addition of four new moderators and the removal of two moderators. The new moderators were pattern area, total pattern area, total area of prey surface, and prey shape types, although similar moderators were in the pre-registration such as the number of eyespots (patterns) and diameter of an eyespot (a pattern). These *post-hoc* decisions were taken to refine our initial moderators. We subsequently used them in our meta-regression analyses. We originally intended to include the broad outcome categories of predator avoidance measure as a moderator in the models, but the diversity of reported results made categorisation impossible. Therefore, we did not include it as a moderator. We also collected information on bird diet but decided not to include it. This decision was because six of the seven bird species in our study were omnivores, resulting in a lack of variability needed to detect diet effects in our data (for more details, please see [Results](#)).

Results

Screening outcomes and dataset characteristics

We obtained 270 effect sizes from 33 studies for our analysis. The screening process and reasons for exclusion at the full-text screening stage are summarised in the PRISMA-like flowchart (**Fig. 2a**), with additional details available in **Table S3**, which comprises a list of included/excluded studies. Of the dataset, 68.9% of effect sizes came from eyespot presentation experiments (**Fig. 2b**). The remaining 31.1% of effect sizes came from non-eyespot pattern presentation experiments (**Fig. 2b**). The latter category encompassed various shapes, including circles (71.4%), rectangles (16.7%), diamonds (6.0%), complex patterns (combinations of circles and diamonds; 4.8%), and stripes (1.1%); 93.7% of the control stimuli used in these experiments involved the removal of the pattern used in the treatment stimuli; the remaining stimuli were camouflage patterns (6.3%). Prey shape type used for stimulus presentation varied from real or imitation of a particular butterfly (24.4%) to simply a piece of paper (21.5%) (**Fig. 2b**). The number of pattern shapes varied between studies from one to 11, but in most experiments, they were two (i.e., a pair of shapes; **Fig. 2c**). Additionally, we found that the size of these patterns, both area and maximum diameter/length, exhibited considerable variation across studies (**Fig. 2c**). The total area of the patterns and stimulus also varied widely (**Fig. 2c**). The studies reported responses to conspicuous pattern stimuli by seven bird species (**Fig. 2d**). Chickens (*Gallus gallus*) and common starlings (*Sturnus vulgaris*) were the most studied birds in our dataset. Apart from chickens (eight studies) and Eurasian blue tits (*Cyanistes caeruleus*; five studies), effect sizes were available from just one or two studies per species. Six of the seven species were omnivores, and one (yellow bunting; *Emberiza sulphurata*) was a granivore [55].

Does the presence of conspicuous patterns affect predator avoidance?

The overall mean effect size was statistically significant, showing a 21.86% (this percentage value is the back-transformed values of $\ln RR$) increase in the probability of predator avoidance, such as higher prey survival rates or eliciting fewer attacks from birds (estimate = 0.20, 95% CI = [0.08, 0.31], $t_{[df=268]} = 3.40$, $p = 0.0008$), in prey with conspicuous patterns than in prey without such patterns (**Fig. 3a**). Total heterogeneity across effect sizes was high ($I^2 = 96.50\%$); more specifically, observation ID (representing the within-study effect) accounted for the most heterogeneity, 79.88%, with study ID (representing between-study effect) accounting for the remaining 16.61%.

Is there a difference in predator avoidance between eyespots and conspicuous patterns?

There was no statistically significant difference between the effects of eyespots and non-eyespot patterns ($F_{[df1=1, df2=268]} = 0.33$, $p = 0.57$, $R^2 = 0.27\%$; **Fig. 3b**). On average, eyespot patterns resulted in 24.37% (estimate = 0.22, 95% CI = [0.08, 0.35], $t_{[df=268]} = 3.17$, $p = 0.002$) and non-eyespot patterns in 17.11% (estimate = 0.16, 95% CI = [-0.02, 0.34], $t_{[df=268]} = 1.71$, $p = 0.09$) increases in predator avoidance compared with control stimuli, although this trend was not statistically significant for non-eyespots (**Fig. 3b**).

What factors promote predator avoidance?

Our uni-moderator meta-regression model with pattern area (individual shape area) showed that larger patterns were associated with an increase in predator avoidance (estimate = 0.11, 95% CI = [0.03, 0.19], $t_{[df=268]} = 2.71$, $p = 0.007$, $R^2 = 8.56\%$; **Fig. 4a**). The total pattern area also promoted

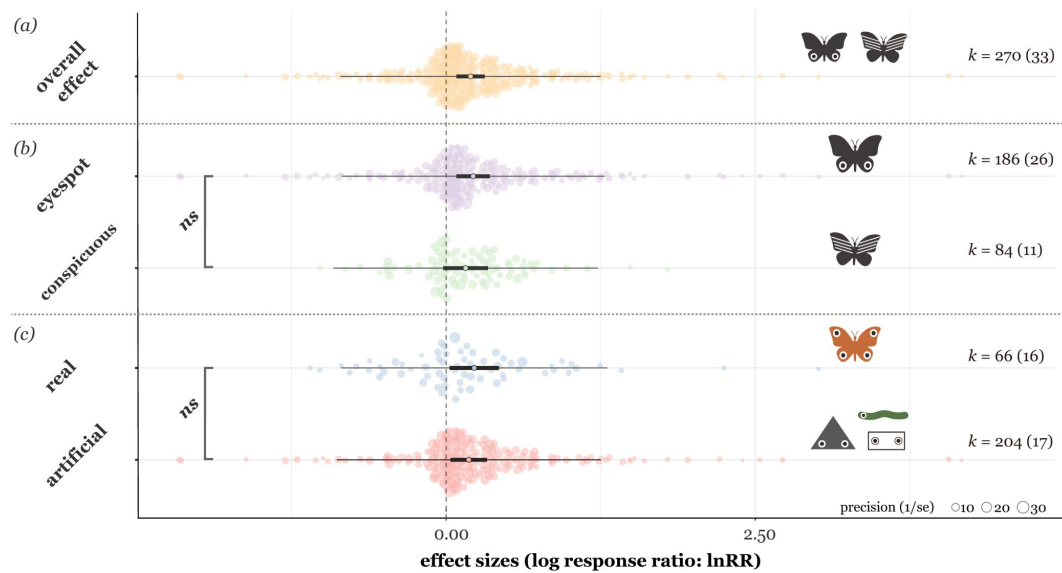


Figure 3

Mean effect sizes of (a) overall for all highly salient patterns, (b) effects split by experiments with eyespots versus conspicuous patterns, and (c) two prey types used in the experiments.

Thick horizontal lines represent 95% confidence intervals, and thin horizontal lines represent 95% prediction intervals. The points in the centre of each thick line indicate the average effect size. *k* is the number of effect sizes used to estimate the statistics, followed by the number of studies in the brackets.

predator avoidance (estimate = 0.09, 95% CI = [0.004, 0.17], $t_{[df=268]} = 2.07$, $p = 0.04$, $R^2 = 5.18\%$; **Fig. S1a**). Similarly, the maximum diameter/length of the pattern positively influenced predator avoidance (estimate = 0.19, 95% CI = [0.04, 0.35], $t_{[df=268]} = 2.46$, $p = 0.01$, $R^2 = 6.62\%$; **Fig. S1b**). In contrast, an increased number of pattern shapes significantly reduced the effect of predator avoidance (estimate = -0.06, 95% CI = [-0.11, -0.008], $t_{[df=268]} = -2.29$, $p = 0.02$, $R^2 = 2.46\%$; **Fig. 4b**). We found no significant effects of total prey surface area on predator avoidance (estimate = -0.03, 95% CI = [-0.15, 0.09], $t_{[df=268]} = -0.48$, $p = 0.63$, $R^2 = 0.42\%$; **Fig. S1c**). Predator avoidance was not statistically significantly affected by differences in whether the presented prey looked like a real lepidopteran species ($F_{[df1=1, df2=268]} = 0.12$, $p = 0.72$, $R^2 = 0.13\%$). Both types of prey material (real/imitation and abstract butterfly) had similar positive trends (**Fig. 3c**), with the former increasing predator avoidance by 25.55% (estimate = 0.23, 95% CI = [0.03, 0.43], $t_{[df=268]} = 2.24$, $p = 0.03$) and the latter by 20.07% (estimate = 0.18, 95% CI = [0.04, 0.33], $t_{[df=268]} = 2.44$, $p = 0.02$). Further, when also considering prey type (**Fig. S2**), abstract and real butterflies significantly exhibited increased predator avoidance by 37.98% (estimate = 0.32, 95% CI = [0.11, 0.53], $t_{[df=268]} = 3.04$, $p = 0.003$) and by 25.40% (estimate = 0.23, 95% CI = [0.03, 0.42], $t_{[df=268]} = 2.25$, $p = 0.03$), respectively, but artificial abstract caterpillars (estimate = 0.07, 95% CI = [-0.18, 0.31], $t_{[df=266]} = 0.53$, $p = 0.60$) and artificial abstract prey (estimate = 0.01, 95% CI = [-0.35, 0.37], $t_{[df=266]} = 0.06$, $p = 0.95$) did not, respectively. When comparing each prey type (e.g., abstract butterfly vs. real butterfly), none of the differences was statistically significant (**Fig. S2**).

The multi-moderator (full) regression model showed that only pattern area positively affected predator avoidance (estimate = 0.10, 95% CI = [0.009, 0.18], $t_{[df=266]} = 2.16$, $p = 0.03$; **Table S4**). Contrary to the uni-moderator regression model, the number of patterns showed no significant effects on predator avoidance, although the consistent trend remained (estimate = -0.05, 95% CI = [-0.11, 0.004], $t_{[df=266]} = -1.84$, $p = 0.07$; **Table S4**). The full model accounted for 8.33% of the variation in the dataset. The complete output of the multi-moderator model is displayed in **Table S4**.

Publication bias

The funnel plot showed no visual sign of funnel asymmetry (**Fig. 5a**). The meta-regression analysis, which included the square root of the inverse of the effective sample size, further supported this observation by showing that the effective sample size did not significantly predict the effect size values (estimate = -0.09, 95% CI = [-0.83, 0.65], $t_{[df=266]} = -0.24$, $p = 0.81$; **Fig. 5b**). There was no detectable trend suggesting that more recent publications consistently showed lower or higher effect size values, which would have indicated the presence of time-lag publication bias (estimate = -0.0008; 95% CI = [-0.01, 0.01], $t_{[df=266]} = -0.12$, $p = 0.90$; **Fig. 5c**). We obtained the same trends from multi-moderator meta-regressions (**Fig. S3**).

Discussion

Eyespots and non-eyespot patterns did not differ significantly in the magnitude of deterring effects (**Fig. 3b**). Avian predators showed similar avoidance responses to the conspicuous patterns compared to control ones (**Fig. 3a**). Specifically, larger pattern sizes played a crucial role in eliciting negative responses from birds (**Fig. 4a**). Further, negative responses from birds showed the tendency to decline with increasing pattern number: single patterns were likely more intimidating than a group of patterns (**Fig. 4b**). Taken together, our results support the conspicuousness hypothesis rather than the eye mimicry hypothesis.

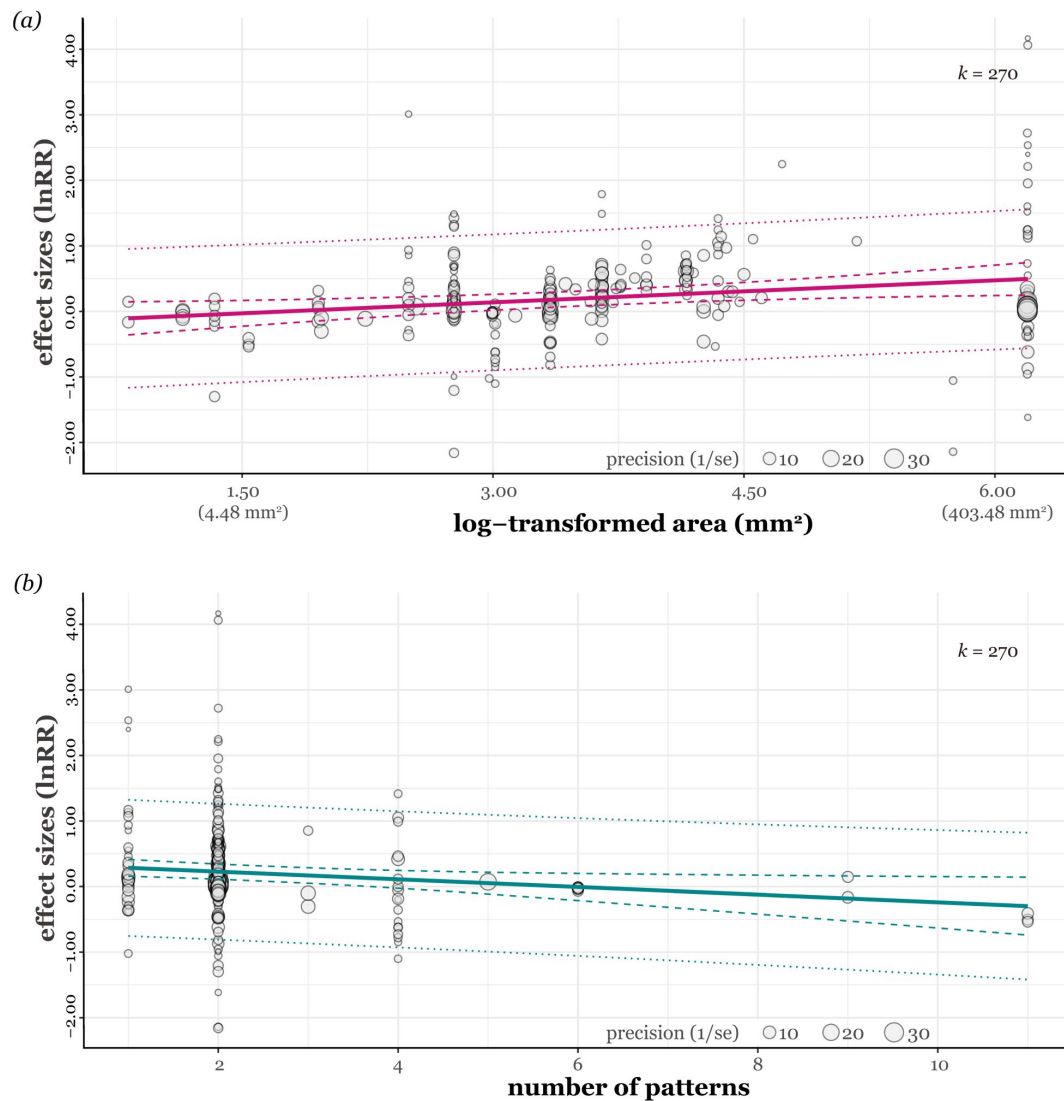


Figure 4

The relationships between (a) prey salient pattern area (log-transformed) and effect sizes and (b) number of prey salient patterns and effect sizes.

Circle sizes are scaled according to precision, k represents the number of effect sizes. Each fitted regression line is shown as a coloured straight line, and 95% confidence and prediction intervals are shown as dashed and dotted coloured lines, respectively.

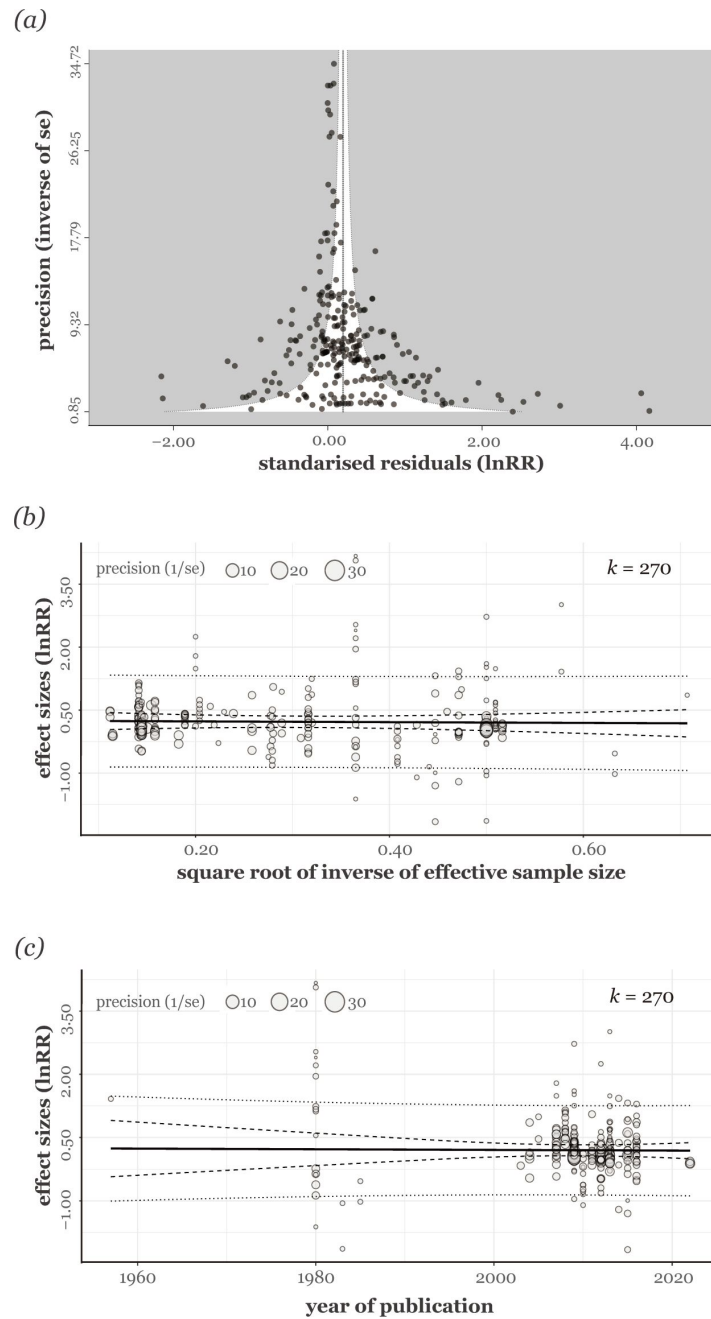


Figure 5

(a) Funnel plot using effect size and its inverse standard error. The relationship between effect sizes and (b) the square root of the inverse of effective sample size and (c) publication year. In (b) and (c), circle sizes are scaled according to precision, and k represents the number of effect sizes. Each fitted regression line is shown as a straight line, and 95% confidence and prediction intervals are shown as dashed and dotted lines, respectively.

Eye mimicry or conspicuous hypothesis?

Overall, our meta-analysis showed that conspicuous patterns could increase predator avoidance by over 20%. Specifically, our results indicate that conspicuousness per se can be advantageous in avoiding bird predation (Fig. 3ab, Fig. 4). The evidence favouring the conspicuousness hypothesis comes mainly from a series of field experiments by Steven and his colleagues [30, 31, 34]. They showed that both eyespots and non-eyesspots improved the prey survival similarly compared to non-conspicuous patterns [30, 31, 34]. In addition, their research showed prey with more conspicuous patterns (i.e., large-size patterns) tended to survive more than others [30, 31, 34], and eye resemblance (e.g., number or pattern shapes) did not significantly affect the prey's survival [30, 31, 34]. Given that these pattern stimuli used in the experiments are rarely or never found in natural environments [34], the most parsimonious explanation for these results is neophobia or dietary conservatism in birds [56–58]. Both phenomena appear to diminish with habituation and/or learning. A few studies investigated such factors for intimidating effects, and they showed that repeated encounters made birds more habituated to eyespot patterns [16, 59, 60]. We need more systematic tests of bird habituation to aposematic-coloured patterns to better understand the evolution and function of such patterns in Lepidoptera.

While our meta-analytic results favour the conspicuousness hypothesis, several empirical studies support the eye mimicry hypothesis. For example, De Bona et al. [28] found that a pair of eyespots of *Caligo martia* was as effective as true owl eyes and more efficient in eliciting predator avoidance responses than less mimetic but equally contrasting circles. Blut and Luau [61] created artificial eye-spotted prey with different similarities to the vertebrate eyes and checked their survival rates in a field experiment. They revealed that the prey with the most mimetic pattern had the highest survival rate [61]. Although studies on Lepidoptera larvae are relatively limited, caterpillar eyespots are considered part of snake mimicry [14]. Some research examined the benefit of eyespots by presenting artificial caterpillars (marked with eyespots and control) made from dyed pastry to wild birds and showed that eyespots improved survival [60, 62, 63]. Despite these convincing pieces of empirical evidence, our meta-analytic results showed that eye resemblance did not improve predator avoidance. If the eye mimic hypothesis was true, we would have seen a clear difference between studies investigating eyespots and non-eyesspots.

However, we observed little heterogeneity among studies, despite finding high heterogeneity within individual studies. This finding implies that if each study followed similar experimental procedures within studies, our main result on predator avoidance would be more generalisable. The high within-study heterogeneity can be caused by varying stimulus characteristics contributing to the effect size variations, even in the same studies. Bird phylogenetic relatedness explained little heterogeneity in our predator dataset, but this may have occurred because a limited number of subject bird species (i.e., chickens, common starlings, Eurasian blue tits) dominated our dataset (Fig. 2d). While we cannot exclude the possibility of species differences in birds' responses to the conspicuous patterns, our analysis indicated that bird species identity did not explain the observed variation in predator avoidance.

We also note that conspicuous patterns can also be important for conspecific communication in butterflies [12, 64]. Eyespots on *Bicyclus anynana* are known to function as sexual signals. For example, males choose females depending on eyespot size and reflectance [65]. Regarding the non-eyespot patterns, males of *Heliconius cydno* and *H. pachinus* can recognise conspecific females by the bright colour of wing patches [66, 67]. Conspicuous patterns can also act as social signals in other taxa (e.g., birds: [68]), but this function remains unclear in butterflies. Therefore, the diversity of patterns on wings could be shaped by intra-specific and inter-specific

communication. We should simultaneously consider the influence of anti-predator and sexual/social signalling functions on the evolution of butterfly conspicuous patterns [cf., 65, 69, 70].

What factors explain the observed heterogeneity?

The indicators of pattern size, including each pattern area (Fig. 4a), total pattern area (Fig. S1a), and maximum diameter/length (Fig. S1b), were the most important moderators of effect sizes, overall indicating that large patterns could promote predator avoidance. Notably, these size metrics were correlated, so they are not independent of each other. Several studies suggested that the pattern size difference is related to the difference in prey survival [21, 26, 30]. For example, eyespots larger than 6.0 mm may have a strong deterrent effect with increasing size [26], but such patterns may increase the visibility of lepidopterans, and their presence may increase predation rates as well [71]. Indeed, small conspicuous patterns tend to attract predators' attention, as explained by the deflection hypothesis [12, 72]. The effect may contribute to the observed negative overall effect sizes (Fig. 3, Fig. 4). Considering studies on *B. anynana* with eyespots with a deflecting effect (maximum diameter is about 5.0 mm; Table S5), a size of at least 6.0 mm is required to avoid predator approach. However, it is uncertain whether the effect would linearly increase with size or whether an optimal size exists. Although eyespot sizes on actual Lepidoptera may be restricted by their body or wing size (e.g., 73), but see also [21]), it would be interesting to find a maximum threshold for patterns that promote predator avoidance responses in birds.

Among other moderators tested (prey material type, total pattern area, and prey shape type), the only moderator that seemed to explain heterogeneity was the number of patterns (Fig. 4b; yet it is likely inconclusive; see Table S4). Previous studies predominantly employed a single pattern or a pair of patterns, leading to limited variations. Nonetheless, our findings indicate that a single eyespot is equally or more effective than a pair of eyespots. Consequently, the resemblance to a pair of eyes, a crucial aspect of the eye mimicry hypothesis, may not be essential for effective predator avoidance. To disentangle the two hypotheses, we recommend conducting the following experiments with two key features [30, 35, 74]: a set of stimuli that (1) have the same size (area or diameter/maximum length of each pattern or total pattern area) but with different numbers of patterns ranging from a few usually found in Lepidoptera to numerous patterns unlike those seen in them, and (2) are presented with the same number of patterns and the same size but different pattern shapes. Results from these experiments could deepen our current knowledge, allowing us to inch toward a more definitive answer.

Knowledge gaps and future opportunities

Along with other conspicuous patterns, eyespots are believed to deter bird predation, and our meta-analysis supports this function. However, five major gaps remain in the current literature and our knowledge. First, birds and humans likely perceive eye-like shapes differently based on the interspecific diversity of bird vision [75]. Thus, it could be premature to conclude that eyespots on Lepidoptera resemble vertebrate eyes universally. For example, most bird species can detect ultraviolet light, which is invisible to humans, and the ultraviolet reflection of the butterflies' eyespots may contribute to predator avoidance [e.g., 20, 22].

Second, some lepidopterans present conspicuous patterns to potential predators in combination with other elements, such as sounds and movements [13, 16, 17, 76, 77], presumably to emphasise the conspicuousness of the patterns. Most of the current literature does not take these effects into account in experiments, although some studies argue in favour or against their importance [e.g., 16, 17]. We should also consider how factors other than those constituting the pattern (e.g., colour, number, and size) are involved in the predator avoidance function of eyespots. The location of the butterfly's eyespot patterns varies from species to species as well; eyespots exist on the wings' ventral, dorsal, or both sides. Not only the dorsal eyespot patterns,

which were used in most studies, but also the ventral eyespot patterns should be explored. In addition, we need to avoid presenting patterns unnaturally when using real butterflies in experiments. For example, many owl butterflies (family *Caligo*) have a pair of eyespot patterns on the ventral side. Their eyespots are usually visible to birds when the wings are closed and would not present side by side as in the eyes of the owl's frontal face.

Third, recent studies have shown that birds are sensitive to the gaze of other individuals and may respond more aversively when their gazes are directed at them [e.g., 78–80]. Skelhorn and Rowland [81] showed that the anti-predation effect may be further enhanced if the inner circle of the eyespot is in a more gazing-like position for subject birds. However, further research is needed to investigate the importance of the position of the inner circle.

Fourth, as mentioned above, studies focusing on caterpillar eyespots are much more scarce compared to butterflies; Hossie and Sherratt [82] have shown similarities between caterpillars and snakes, but the response of birds to actual caterpillars has not been experimentally tested. Conversely, in butterflies, similarities between the eyespot patterns on wings and the eyes of birds of prey have not been investigated.

Finally, birds are generally considered as potential predators of butterflies and caterpillars. Although other taxa species, such as invertebrates [83–85], lizards [27, 86, 87], and rats [88–91], are also known to prey on lepidopterans, there are much fewer studies using non-avian species as predators. Therefore, we should expand the range of taxa used for experiments to get a better and more generalisable understanding of the eyespots' function and evolution in butterflies and caterpillars.

Knowing the effects of conspicuous patterns may contribute to creating a world where birds and humans can live more harmoniously. Both eyespots and conspicuous patterns have already been used to control birds, particularly in agriculture, although their effectiveness has been questioned [e.g., 92, 93]. Such uncertainty may reflect our limited understanding of why birds avoid eyespots and conspicuous patterns. Nevertheless, visual stimuli are less likely to harm birds or affect the natural environment than others (e.g., nest/egg destructions or toxic chemicals; reviewed in [94]). Therefore, when proven effective, they could be used for better pest control, population management and conservation [95].

Conclusion

We have shed light on a traditional but controversial research topic that has fascinated behavioural ecologists for decades. Our findings provide a better understanding of the evolution of signal designs, but also show that more work is needed to understand the function of the eyespot patterns in *Lepidoptera*, such as whether eyespot patterns evolved due to mimicry or conspicuousness.

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Author contribution

AM: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing

ML: Data curation, Funding acquisition, Investigation, Methodology, Writing – review & editing

PP: Data curation, Investigation, Writing – review & editing

YY: Data curation, Investigation, Writing – review & editing

MS: Conceptualization, Funding acquisition, Visualization, Writing – review & editing

SN: Conceptualization, Funding acquisition, Methodology, Software, Supervision, Project administration, Writing – review & editing

All gave final approval for publication and agreed to be held accountable for the work performed therein.

Data Accessibility

Raw data, analysis script and supplementary materials are available at <https://ayumi-495.github.io/eyespot/> . Once the paper is published, these will all be uploaded to Zenodo.

Supplementary figures and tables

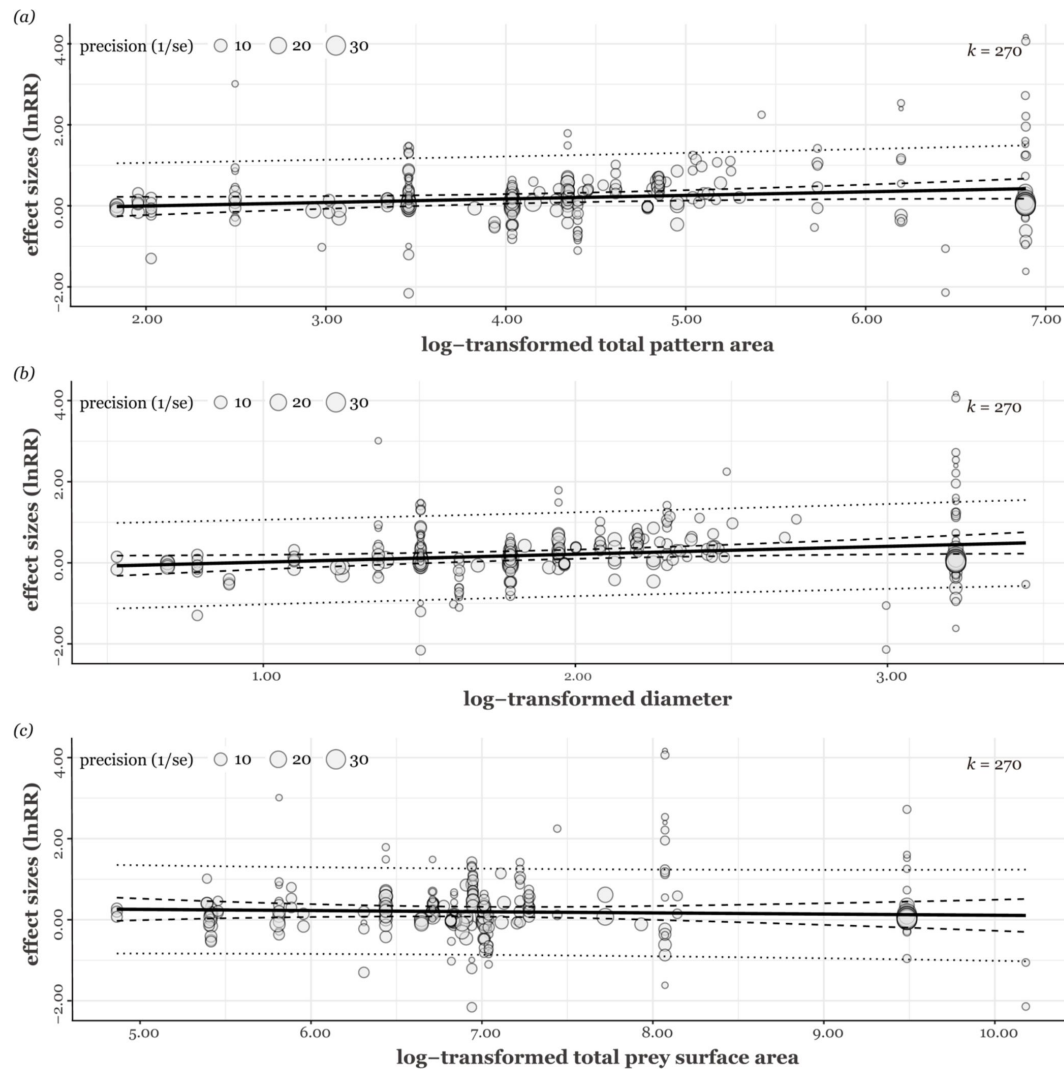


Figure S1

The relationships between (a) total pattern area, (b) pattern maximum diameter/length, and (c) total prey surface area and effect sizes.

k shows the number of effect sizes. Each fitted regression line is shown as a solid straight line, and 95% confidence and prediction intervals are shown as dashed and dotted lines, respectively.

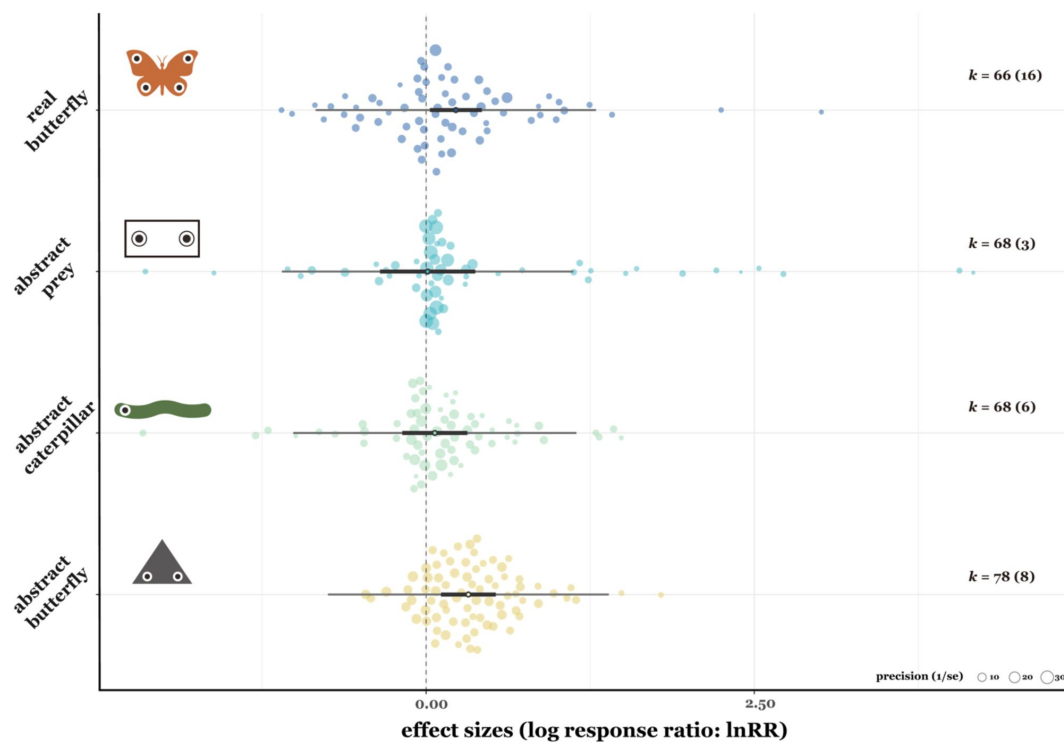


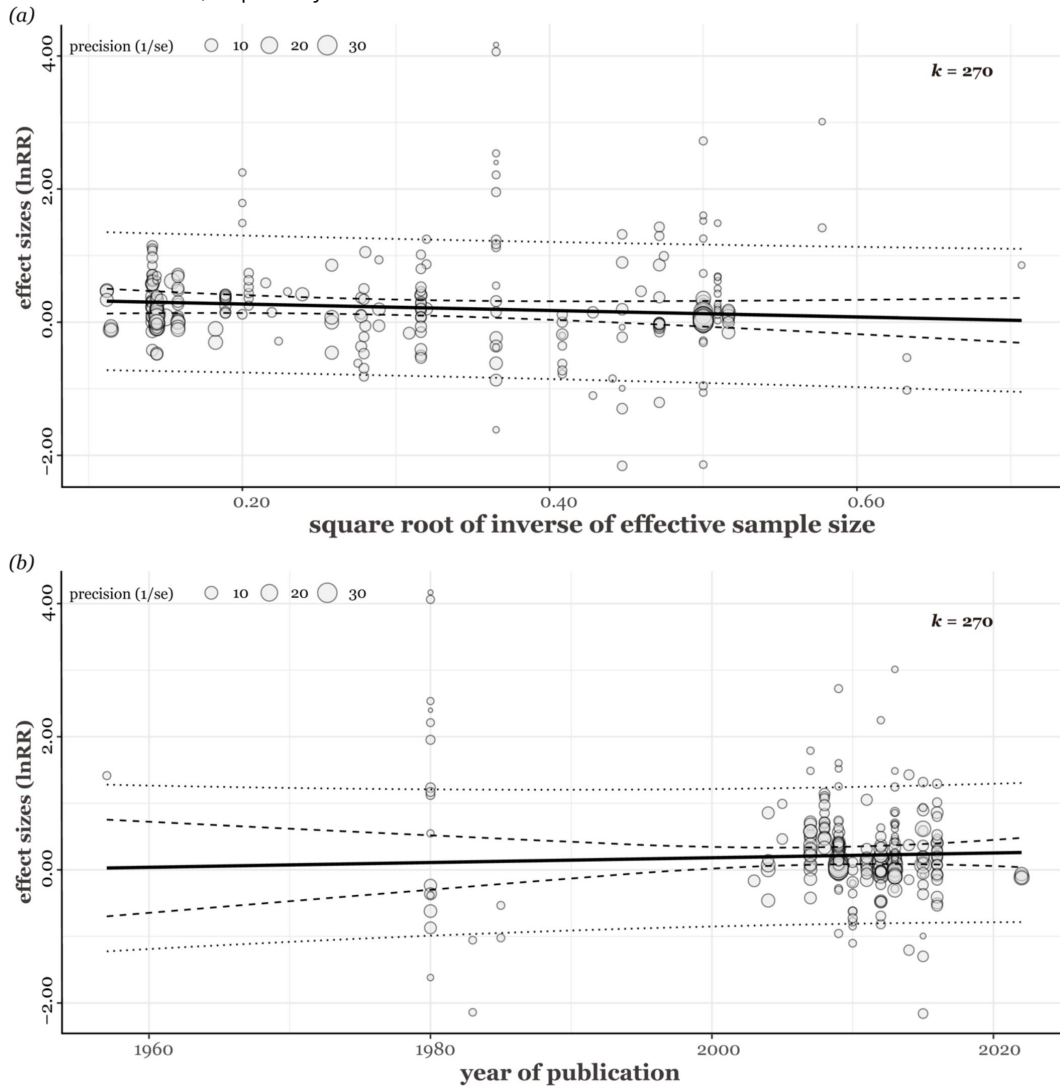
Figure S2

Mean effect sizes of total prey shape types.

Thick horizontal lines represent 95% confidence intervals, and thin horizontal lines represent prediction intervals. The points in the centre of each thick line indicate the average effect size. k shows the number of effect sizes.

Figure S3

(a) relationship between effect size and the square root of the inverse of effective sample size, and (b) relationship between effect size and publication year. Both plots were based on the multi-moderator model. k shows the number of effect sizes. Each fitted regression line is shown as a solid straight line, and 95% confidence intervals and prediction intervals are shown as dashed and dotted lines, respectively.



Database	Search strings
Scopus	TITLE-ABS-KEY (((eyespot* OR eye-spot* OR "eye spot*" OR eye-like* OR "eye like*" OR eye-mimic* OR "eye mimic*" OR "eye similari*" OR "predator* eye*" OR "eye similar*" OR concentric*) AND (attack* OR antipredator* OR anti-predator* OR aposematic* OR avoid* OR conspicuous* OR warn* OR fear* OR intimidat* OR predator-prey* OR butterfly* OR moth* OR bird* OR avian* OR caterpillar* OR prevent* OR aves OR passeri*)) AND NOT (fish* OR manti* OR lizard* OR bat* OR nano* OR health* OR patients OR women OR men OR children OR pediatric OR medic* OR hormon* OR genes OR magnet* OR valve* OR fluid* OR concrete OR beam* OR tissue* OR charge* OR energ* OR electro*))
ISI Web of Science	TS = (((eyespot* OR eye-spot* OR "eye spot*" OR eye-like* OR "eye like*" OR eye-mimic* OR "eye mimic*" OR "eye similari*" OR "predator* eye*" OR "eye similar*" OR concentric*) AND (attack* OR antipredator* OR anti-predator* OR aposematic* OR avoid* OR conspicuous* OR warn* OR fear* OR intimidat* OR predator-prey* OR butterfly* OR moth* OR bird* OR avian* OR caterpillar* OR prevent* OR aves OR passeri*)) NOT (fish* OR manti* OR lizard* OR bat* OR nano* OR health* OR patients OR women OR men OR children OR pediatric OR medic* OR hormon* OR genes OR magnet* OR valve* OR fluid* OR concrete OR beam* OR tissue* OR charge* OR energ* OR electro*))
BASE	eyespot* AND (avoid* predator* prevent* intimidat* mimi*) AND (ave* bird* passerine* butterfly* moth* lepidoptera caterpillar*) AND (experiment* stud*)
Google scholar	eyespot avoid predator prevention intimidation mimic aves bird passerine butterfly moth lepidoptera caterpillar experiment study
	We translated the above English search string into <i>Japanese, Polish, Portuguese, Russian, Spanish, Simplified Chinese, and Traditional Chinese</i> for searching on Google Scholar. <i>Japanese:</i> 目玉模様 眼状紋 忌避 捕食 防除 威嚇 擬態 鳥 鳴禽 蝶 蛾 鱗翅目 芋虫 幼虫 実験 研究 <i>Polish:</i> oko oczy skrzydła wzor plama ochrona unikanie drapieźnik zapobieganie zastraszenie ptak motyl gasienica owad eksperyment badania <i>Portuguese:</i> ocelo "mancha ocelar" "olhos falsos" "falsos olhos" evitar predador prevenção intimidação ave pássaro borboleta mariposa lagarta experimento estudo <i>Russian:</i> глаз глаза избегать хищник профилактика запугивание птица бабочка мотылек Воробыинообразные Чешуекрылые Гусеница эксперимент изучать <i>Spanish:</i> ocelo "ojos falsos" "falsos ojos" evitar depredador prevención intimidación ave pájaro mariposa polilla oruga experimento estudio <i>Simplified chinese:</i> 眼点 避免 捕食者 预防 恐吓 模仿 鸟类 鸟 雀 蝴蝶 蛾 鳞翅目 毛毛虫 实验 试验 学习 <i>Traditional chinese:</i> 眼點 避免 捕食者 預防 恐嚇 模仿 鳥類 鳥 雀 蝴蝶 蛾 鱗翅目 毛毛蟲 實驗 試驗 學習

Table S2

Search strings used for each database.

We accessed Scopus, ISI Web of Science core collection, Google Scholar (*Japanese, Polish, Portuguese, Russian, Spanish, Simplified Chinese, and Traditional Chinese*) on 08/06/2023, and Bielefeld Academic Search Engine (BASE) on 26/06/2023. BASE was used as a source of grey literature. We conducted backward and forward reference searches for key review articles using Scopus on 19/06/2023. We modified search strings to collect studies to capture studies examining the effects of eyespot patterns on birds using experimental methods. Search strings were adapted to the structure of each database.

(a) included studies					
title	year	authors	journal	doi	
The function of Eyespot Patterns in the Lepidoptera	1957	Blair, AD.	Behaviour	10.1163/156853956X300048	
Reactions of male domestic chicks to two-dimensional eye-like shapes	1980	Jones, RB.	Animal Behaviour	10.1016/S0003-3472(80)80025-X	
The Feeding Behaviour of Starlings (<i>Hirundo vulgaris</i>) in the Presence of 'Eyes'	1983	Inglis, RL, Hixon, LW, Marshall, MB, and Neville, PA.	Zeitschrift für Tierpsychologie	10.1111/j.1439-0310.1983.tb02151.x	
Butterfly wing markings are more advantageous during handling than during the initial strike of an avian predator	1985	Woums, MK, and Wasserman, FE.	Evolution	10.1111/j.1558-5646.1985.tb00426.x	
Significance of butterfly eyespots as an anti-predator device in ground-based and aerial attacks	2003	Lysitnes, A., Brakefield, PM, and Mappes, J.	Oikos	10.1034/1600-0706.2003.11935.x	
Does predation maintain eyespot plasticity in <i>Bicyclus anynana</i> ?	2004	Lysitnes, A., Brakefield, PM, Lindström, L., and Mappes, J.	Proceedings of the Royal Society B: Biological Sciences	10.1098/rspb.2003.2571	
Asymmetry in size, shape, and color impairs the protective value of conspicuous color patterns	2004	Foranias, A, and Herberstein, J.	Behavioral Ecology	10.1093/beheco/arg092	
Prey survival by predator intimidation: an experimental study of peacock butterfly defense against blue tits	2005	Vallin, A, Jakobsson, S., Lind, J, and Wiklund, C.	Proceedings of the Royal Society B: Biological Sciences	10.1098/rspb.2004.0154	
Field experiments on the effectiveness of 'eyespot' as predator deterrents	2007	Stevens, M., Hopkins, E., Hinde, W., Adcock, A., Connolly, Y., Trzcinski, T. and Cuthill, IC.	Animal Behaviour	10.1016/j.anbehav.2007.01.031	
The anti-predator function of 'eyespot' on camouflaged and conspicuous prey	2008	Stevens, M., Stubbin, CL, and Hardman, CJ.	Behavioral Ecology and Sociobiology	10.1007/s00265-008-0607-3	
Conspicuousness, not eye mimicry, makes "eyespot" effective antipredator signals	2008	Stevens, M., Hardman, CJ, and Stubbin, CL.	Behavioral Ecology	10.1093/beheco/arn162	
The protective value of conspicuous signals is not impaired by shape, size, or position asymmetry	2009	Stevens, M., Catter, Perry, SA, and Price, JRF.	Behavioral Ecology	10.1093/beheco/arn119	
The function of animal 'eyespot': Conspicuousness but not eye mimicry is key	2009	Stevens, M., Catter, A., Grahame, J, and Winney, JS.	Current Zoology	10.1093/czoolo/55.5.319	
Fixed eyespot display in a butterfly thwarts attacking birds	2009	Kodandaramiah, U., Vallin, A, and Wiklund, C.	Animal Behaviour	10.1016/j.anbehav.2009.02.018	
Can we use starlings' overreaction to eyespots as the basis for a novel 'cognitive blur' task?	2009	Briot, BR., Normandale, CL., Parkin, A, and Bateson, M.	Applied Animal Behaviour Science	2009.02.015	
Constant eyespot display as a primary defence-survival of male and female emperor moths attacked by blue tits	2010	Vallin, A., Sven J. and Christer W.	The Journal of Research on the Lepidoptera	10.5962/p.266504	
Defensive effect and the effect of prey detectability on anti-predator function of eyespots	2011	Vallin, A, and Diminova, M., Kodandaramiah, U, and Meriläin, S.	Behavioral Ecology and Sociobiology	10.1007/s00265-011-1173-7	
Number of eyespots and their intimidating effect on naïve predators in the peacock butterfly	2011	Meriläin, S., Vallin, A., Kodandaramiah, U., Diminova, M., Ruuskanen, S, and Laaksonen, T.	Behavioral Ecology	10.1093/beheco/arn135	
The 'sparkle' in fake eyes - the protective effect of mimic eyespots in lepidoptera	2012	Blut, C., Wilbrandt, J., Fels, D, Gügel, Experimentalia et Applicata	Entomologia Experimentalis et Applicata	10.1111/j.1365-3113.2012.01260.x	
Eyespots interact with body colour to protect caterpillar-like prey from avian predators	2012	Hosie, TJ, and Sherratt, TN.	Animal Behaviour	10.1016/j.anbehav.2012.04.027	
Anti-predator adaptations and strategies in the Lepidoptera	2012	de Wirt, L.	Doctoral thesis	none	
Bird attacks on a butterfly with marginal eyespots and the role of prey concealment against the background	2013	Olofsson, M., Jakobsson, S, and Wiklund, C.	Biological Journal of the Linnean Society	10.1111/bj.12063	
Defensive posture and eyespots deter avian predators from attacking caterpillar models	2013	Hosie, TJ and Sherratt, TN.	Animal Behaviour	10.1016/j.anbehav.2013.05.029	
Revealed by conspicuousness: distractive markings reduce camouflage	2013	Stevens, M., Marshall, KLA, Trzcinski, J., Pailley, S., Burnand, D, and Chadwick, M.	Behavioral Ecology	10.1093/beheco/arn156	
Eyespot display in the peacock butterfly triggers antipredator behaviors in naïve adult fowl	2013	Olofsson, M., Lovlie, H., Tibblin, J., Jakobsson, S, and Wiklund, C.	Behavioral Ecology	10.1093/beheco/arn167	
The position of eyespots and thickened segments influence their protective value to caterpillars	2014	Skelhorn, J., Derrington, G., Hosie, TJ, and Sherratt, TN.	Behavioral Ecology	10.1093/beheco/arn154	
Predator mimicry, not conspicuousness, explains the efficacy of butterfly eyespots	2015	De Bous, S., Valkonen, R., López-Sepulcre, A, and Mappes, J.	Proceedings of the Royal Society B: Biological Sciences	10.1098/rspb.2015.0202	
Body size affects the evolution of eyespots in caterpillars	2015	Hosie, TJ, Skelhorn, J., Beinstadt, JW., Kawahara, AY, and Sherratt, TN.	Proceedings of the National Academy of Sciences of the United States of America	10.1073/pnas.1415121112	
What makes eyespots intimidating- the importance of pureness Evolutionary ecology and behaviour	2015	Mukherjee, R, and Kodandaramiah, U.	BMC Evolutionary Biology	10.1186/s12862-015-0307-3	
On the deterring effect of a butterfly's eyespot in juvenile and sub-adult chickens	2015	Olofsson, M., Wiklund, C, and Favati, A.	Current Zoology	10.1093/czoolo/61.4.749	
Multi-component deceptive signals reduce the speed at which predators learn that prey are profitable	2016	Skelhorn, J., Holms, CG., Hosie, TJ, and Sherratt, TN.	Behavioral Ecology	10.1093/beheco/arn135	
Attack risk for butterflies changes with eyespot number and size	2016	Hö, S., Schuchat, SK., Piel, WH, and Montello, A.	Royal Society Open Science	10.1098/rsos.150614	
The effectiveness of eyespots and masquerade in protecting artificial prey across ontogenetic and seasonal shifts	2022	Postema, EG.	Current Zoology	10.1093/cz/znab082	

Table S3

List of (a) included and (b) excluded studies at the full-text screening stage with exclusion reasons.

(b) excluded studies

title	year	authors	journal	doi	reason
The effects of a tranquilliser on the reactions of domestic chicks to an aversive eye-like shape	1979	Jones, RB.	IRCS Medical Science	none	No full-text
Young domestic chicks avoid eye-like shapes	1980	JONES, RB	Applied Animal Ethology	10.1016/0304-3762(80)90037-1	No full-text
The startle responses of blue jays to Catocala	1985	Schlenoff, DH.	Animal Behaviour	10.1016/S0003-3472(85)80164-0	Wrong outcome
<i>(Lepidoptera: Noctuidae)</i> prey models					
Fearful symmetry: Pattern size and asymmetry affects aposematic signal efficacy	1999	Forsman, A. and Merilaita, S.	Evolutionary Ecology	10.1023/A:1006630911975	Invalid comparator
"An eye for an eye?" - On the generality of the intimidating quality of eyespots in a butterfly and a hawkmoth	2007	Vallin, A., Jakobsson, S. and Wiklund, C.	Behavioral Ecology and Sociobiology	10.1007/s00265-007-0374-6	Invalid comparator
Coincident disruptive coloration	2009	Cuthill, IC and Szekeley, A	Philosophical Transactions of the Royal Society B-Biological Science	10.1098/rstb.2008.0266	Invalid comparator
Marginal eyespots on butterfly wings deflect bird attacks under low light intensities with UV wavelengths	2010	Olofsson, M., Vallin, A., Jakobsson, S. and Wiklund, C.	PLoS ONE	10.1371/journal.pone.0010798	Invalid comparator
Insect coloration as a defence mechanism against visually hunting predators	2011	Lyytinen, A.	Doctoral thesis	none	Published thesis
Effects of lepidopteran eyespot components on the deterrence of predatory birds	2015	Blut, C. and Lunau, K.	Behaviour	10.1163/1568539X-00003288	Invalid comparator
Antipredator behavior by a nesting hummingbird in response to a caterpillar with eyespots	2019	Marden, JH. and Pérez Carrillo, JF.	Ecology	10.1002/ecy.2582	Observational study
The Influence of the eyespots of peacock butterfly (<i>Aglais io</i>) and caterpillar on predator recognition	2020	Park, J. and Heo D	Open Science Journal	10.23954/osj.v5i2.2455	Invalid comparator

Table S3 (continued)

Table S4

Summary of a multi-moderator model including all moderators.

The bold typeface is used when a 95% confidence interval (CI) does not contain zero; thus, it can be interpreted as an existing significant effect in predator avoidance.

	Estimate	95%CI
intercept	-0.06	(-0.50, 0.34)
Treatment stimulus	-0.02	(-0.19, 0.23)
Log-transformed area	0.09	(0.009, 0.18)
Number pattern	-0.05	(-0.11, 0.004)
Material type of prey: real	0.18	(-0.09, 0.45)

Table S5

Average maximum diameter of Eyespots on *Bicyclus anynana*.

AM obtained the pictures from lepdata.org/photos/animals/ and <https://data.nhm.ac.uk/> and measured the eyespot diameters. Raw data is <https://ayumi-495.github.io/eyespot/>.

Median	Range
3.41	1.82 – 5.04

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Editors

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Senior Editor

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Pennsylvania State University, University Park, United States of America

Reviewer #1 (Public Review):

Summary:

The question of whether eyespots mimic eyes has certainly been around for a very long time and led to a good deal of debate and contention. This isn't purely an issue of how eyespots work either, but more widely an example of the potential pitfalls of adopting 'just-so-stories' in biology before conducting the appropriate experiments. Recent years have seen a range of studies testing eye mimicry, often purporting to find evidence for or against it, and not always entirely objectively. Thus, the current study is very welcome, rigorously analysing the findings across a suite of papers based on evidence/effect sizes in a meta-analysis.

Strengths:

The work is very well conducted, robust, objective, and makes a range of valuable contributions and conclusions, with an extensive use of literature for the research. I have no issues with the analysis undertaken, just some minor comments on the manuscript. The results and conclusions are compelling. It's probably fair to say that the topic needs more experiments to really reach firm conclusions but the authors do a good job of acknowledging this and highlighting where that future work would be best placed.

Weaknesses:

There are few weaknesses in this work, just some minor amendments to the text for clarity and information.

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Reviewer #2 (Public Review):

Many prey animals have eyespot-like markings (called eyespots) which have been shown in experiments to hinder predation. However, why eyespots are effective against predation has been debated. The authors attempt to use a meta-analytical approach to address the issue of whether eye-mimicry or conspicuousness makes eyespots effective against predation. They

state that their results support the importance of conspicuousness. However, I am not convinced by this.

There have been many experimental studies that have weighed in on the debate. Experiments have included manipulating target eyespot properties to make them more or less conspicuous, or to make them more or less similar to eyes. Each study has used its own set of protocols. Experiments have been done indoors with a single predator species, and outdoors where, presumably, a large number of predator species predated upon targets. The targets (i.e. prey with eyespot-like markings) have varied from simple triangular paper pieces with circles printed on them to real lepidopteran wings. Some studies have suggested that conspicuousness is important and eye-mimicry is ineffective, while other studies have suggested that more eye-like targets are better protected. Therefore, there is no consensus across experiments on the eye-mimicry versus conspicuousness debate.

The authors enter the picture with their meta-analysis. The manuscript is well-written and easy to follow. The meta-analysis appears well-carried out, statistically. Their results suggest that conspicuousness is effective, while eye-mimicry is not. I am not convinced that their meta-analysis provides strong enough evidence for this conclusion. The studies that are part of the meta-analysis are varied in terms of protocols, and no single protocol is necessarily better than another. Support for conspicuousness has come primarily from one research group (as acknowledged by the authors), based on a particular set of protocols.

Furthermore, although conspicuousness is amenable to being quantified, for e.g., using contrast or size of stimuli, assessment of 'similarity to eyes' is inherently subjective. Therefore, manipulation of 'similarity to eyes' in some studies may have been subtle enough that there was no effect.

There are a few experiments that have indeed supported eye-mimicry. The results from experiments so far suggest that both eye-mimicry and conspicuousness are effective, possibly depending on the predator(s). Importantly, conspicuousness can benefit from eye-mimicry, while eye-mimicry can benefit from conspicuousness.

Therefore, I argue that generalizing based on a meta-analysis of a small number of studies that conspicuousness is more important than eye-mimicry is not justified. To summarize, I am not convinced that the current study rules out the importance of eye-mimicry in the evolution of eyespots, although I agree with the authors that conspicuousness is important.

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